Inverse-Lucas correction for a ten-positive consistency equation

Research prepared for Vladimir Reshetnikov, with AI assistance

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1 Inverse-Lucas correction: 85 operations in ten positive variables

Result and scope. The literal polynomial `positive/equation_bV.json` has ten independently strictly positive integer unknowns, exact degree 304, and 85 live binary arithmetic operations. The newest inspected Lucas-correction parent has the same degree and 87 operations. The square-weight control has 84 operations and degree 410; it is dominated by known earlier points and is not promoted as a frontier improvement. The complete indexed BFS943/radix67 tuple is unchanged, with largest fixed port of 592,884 bits.

Composed with the explicitly retained ordinary native/refutation calibration, the proof below gives

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in \mathbb{Z}_{>0}^{10}\ P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

This neither proves consistency nor asserts a zero. It is an ordinary mathematical proof with exact finite artifact checks, not a new end-to-end proof-assistant object or a global-minimum claim.

The new mechanism replaces the forward rank correction by an inverse Lucas correction at the existing abscissa k. This saves two multiplications and removes the exceptional negative-rank transport cases. A genuine obstruction at the prime 3 means that the old denominator proof cannot simply be reused. Projection must precede full denominator recovery; a new absolute bound handles a possibly negative corrected quotient in the positive lift. The second-order expansion and the projection-first strategy themselves are explicitly inherited from the inspected Lucas Corrections and Cofinal Witnesses archives at commit `7a8da414376a80b90ecf799f37ceb26fb9ac674e`.

1.1 Complete polynomial

The only unknowns are

$$J, F, \alpha, i, T, w, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

The fixed integers β, K_0, M_C, M_F are the whole supplied tuple, with $\beta \geq 31$ odd, $K_0 > 0$, and $0 < M_C < \beta < M_F < 2\beta$. Define polynomial abbreviations:

$$\begin{gathered}
h = \beta J, \quad q = h + 1, \quad C = q - F - \alpha, \\
X = qw, \quad Y = qs, \quad k = \eta + \zeta, \quad c = kY + \eta, \\
A = Y(X + 1) + 1, \quad \Delta = A^2 - 1, \quad H = 4A - 5, \\
v = XY^2, \quad \ell = vk = (XY)(kY), \quad E = 1 + \ell(\ell + k), \\
R = [h(q - F) + \alpha](q^2 - 1) + (M_C + qM_F)J, \\
B = (K_0 + w)C - F, \quad U = ic^2, \quad a = \Delta U^2 E, \\
F_0 = 1 + a, \quad G = EF_0, \quad F_s = 1 + 4aF_0, \quad Q = \Delta(F_s - 1), \\
b = cT - R, \quad V = F_s b - c, \quad K = X(3c - 2X) + 2, \\
e_6 = 6(k - R - 1) - \ell(k^2 - 4), \quad D_6 = 6q^6, \quad M = hD_6H.
\end{gathered}$$

Use $L = bV$ for the principal equation, or $L = V^2$ for the control, and put

$$Z = hD_6\big[H(n - LB) + K\big] - H\big[D_6B + h(e_6 - 6q^4 s)\big], \tag{1}$$

$$P = Q(Z^2 - M^2GV^2) + (M^2G - Z^2). \tag{2}$$

The single equation is $P = 0$. All displayed powers have fixed nonnegative integer exponents. The DAG does not compute v separately: it reuses $\ell = (XY)(kY)$. There are no division gates, variable exponents, sequence functions, or additional quantified variables.

For proof only define

$$\begin{gathered}
e_* = \frac{e_6}{6} = k - R - 1 - \frac{vk(k^2 - 4)}{6}, \quad \rho = \frac{e_*}{q^6}, \quad \sigma = \frac{s}{q^2}, \\
t = \frac{B}{h}, \quad W = 1 + hL, \quad \kappa = \frac{K}{H}, \quad z = \frac{Z}{M}.
\end{gathered}$$

Without assuming any integrality of these quotients,

$$z = n + \kappa + \sigma - Wt - \rho. \tag{3}$$

1.2 Root extraction and uniform signed transport

The unchanged root and folding arguments are retained in `inherited_lucas/PROOF.md`, Sections 2–3, with their ordinary Pell sources. We state their premises and conclusions to specify the interface precisely. Positive inputs give $q \geq 32$, $X, Y \geq q$, $k \geq 2$, $c > k$, $A > 4$, $U \geq c^2$, $F_s > 2c$, and $Q > c^2$. Moreover

$$F_s = (1 + 2a)^2, \qquad Q = 4\Delta^2U^2G, \qquad \gcd(E, 1 + a) = 1.$$

At a zero, prime valuations in $(Q - 1)z^2 = G(QV^2 - 1)$ give $G = g^2$, $E = \tau^2$, $F_0 = f_0^2$, with $g = \tau f_0 > 0$ integral. Rational adjacent-Pell descent makes $y = |z|/g$ integral, and

$$QV^2 - (Q - 1)y^2 = 1, \quad z = \varepsilon g y, \quad \varepsilon \in \{1, -1\}.$$

Writing $f = 1 + 2a$, $I = 2Ug$, we have $f^2 - \Delta I^2 = 1$, $Q = (\Delta I)^2$, and $c^2 \mid I$. Signed all-index folding gives

$$c = \psi_A(p) \quad (p > 0 \text{ odd}), \qquad k = 2\psi_{1+2v}(m), \quad m \geq 2,$$

where $\psi_a(0) = 0$, $\psi_a(1) = 1$, and $\psi_a(j + 2) = 2a\psi_a(j + 1) - \psi_a(j)$. In particular $k \geq 8v + 4$, $c > 8XY^3 \geq 8q^4$, $c > X$, and $K > 0$.

The inherited exclusions of main and auxiliary index three use only $q \mid X, Y$, not the rank correction. They give, for either weight,

$$b \neq 0, \quad \operatorname{sign} V = \operatorname{sign} b, \quad |V| > |b|, \quad L > 0, \quad gy < L/2, \quad \frac{|b|}{q^6} < L/2. \tag{4}$$

For $L = bV$, the main index-three remainder would force $2Y - X \in \{1, 2\}$, impossible modulo q. The two signed auxiliary index-three possibilities would give

$$F_s(b - 4\Delta) = c - (4\Delta + 3), \qquad F_s(b + 4\Delta) = c + (4\Delta + 3),$$

whose right sides lie strictly between 0 and F_s. Thus the auxiliary index is at least five, giving $|b| > 4g$ and $gy < 3L/8$. Neither sign branch is discarded.

The new correction has an advantage before any sign of R is known: $k \geq 2$ makes $vk(k^2 - 4)/6 \geq 0$. Therefore

$$e_* \leq k - R - 1 < b, \qquad \rho < |b|/q^6.$$

If $B \leq -1$, then $Wt < -L$, and (3)–(4) force

$$n = \varepsilon gy - \kappa - \sigma + Wt + \rho < L/2 - L + |b|/q^6 < 0,$$

a contradiction. This handles all negative R and $R = 0$ uniformly. Hence $B \geq 0$, so $C > 0$. Only now is the inherited mask bound $3q < R < q^4 - 2$ invoked. Then $c > 2R$ and $b, V > 0$; the normalized-rank and signed auxiliary-index theorems give $p = R$. Thus R is positive odd; mask parity gives even q and odd J.

1.3 Inverse-Lucas authentication at all primes

The inherited, universally valid finite Lucas expansion gives

$$k = 2m + 8vC_m + 32v^2 Z_m, \quad C_m = \binom{m+1}{3}, \quad Z_m \in \mathbb{Z}_{\geq 0}. \tag{5}$$

Binomial notation occurs only in the proof. Since $k = 2u$ is even, $k(k^2-4)/6 = 4u(u-1)(u+1)/3$ is integral. Thus e_* is integral at a zero. Clearing the integral carrier gives $6q^6 \mid hH(e_6 - 6q^4 s)$; cancelling the ordinary factor six and using $h = q - 1$, $H \equiv -1 \pmod{q}$, gives

$$q^6 \mid e_* - q^4 s, \qquad q^4 \mid e_*. \tag{6}$$

No modular inverse of six is used.

Put $t_0 = 2m$, $d = 2m - R - 1$, $A_m = C_m + 4vZ_m$, and $S_m = k^2 + kt_0 + t_0^2 - 4$. The exact inverse identity is

$$3(e_* - d) = 4v^2(24Z_m - A_m S_m). \tag{7}$$

Indeed $t_0(t_0^2 - 4) = 48C_m$, $k - t_0 = 8vA_m$, and $k(k^2 - 4) - t_0(t_0^2 - 4) = (k - t_0)S_m$. Since $v = q^3 w s^2$, the right side is divisible by q^6. For each prime $\lambda \mid q$, with $a_\lambda \geq 1$ its valuation in q,

$$\operatorname{ord}_\lambda(e_* - d) \geq 6a_\lambda - \operatorname{ord}_\lambda(3) \geq 4a_\lambda.$$

Hence $q^4 \mid d$. The first base $1 + 2XY^2$ exceeds A. If $m \geq R$, monotonicity would give $k/2 \geq \psi_A(R) = c$, contrary to $c > k$. Thus $1 \leq m < R$, and $|d| < R < q^4$, proving

$$d = 0, \qquad m = (R + 1)/2. \tag{8}$$

It would be false to infer full q^6-divisibility here. For $q = 3, v = 27, m = 2, R = 3$, the inverse error is $-47{,}911{,}824 \equiv 243 \pmod{729}$. This modular diagnostic is not a purported zero of the full polynomial. Evenness of q alone does not remove the obstruction at three.

1.4 Projection first, denominator second

Because $z = Z/M$ is integral, reducing the literal numerator (1) modulo H gives $H \mid 6hq^6K$. As $\gcd(q, H) = 1$,

$$H \mid 6hK. \tag{9}$$

This uses neither integral ρ nor integral σ, and does not cancel three. Write $R = 2r + 1$, so $r \geq 48$. The retained uniform ratio theorem, licensed by the authenticated indices, gives

$$Y > X^r/3, \qquad X^{r+1} < 3Y(X + 1) < H.$$

Modulo odd H, the scaled recurrence gives $3 \cdot 2^{R-1}\psi_A(R) \equiv 4^R - 1$, because $4A \equiv 5$. Consequently $2^{R-1}K \equiv -(X - 2^R)(2^RX + 1) \pmod H$. Equation (9) therefore implies divisibility of this whole product by H, with multiplier $6h$.

For an explicit height check let $u_0 = 2^R = 2 \cdot 4^r$. Since $6 < q^3$ and $q^3h < X^4$,

$$\begin{aligned} 6h\,|(X - u_0)(u_0X + 1)| &< X^4(X + u_0)(u_0X + 1) \\ &\leq 4 \cdot 4^rX^6 + 8 \cdot 16^rX^5 < X^{r+1} < H. \end{aligned}$$

After division by X^{r+1}, the two terms decrease with $X \geq 32, r \geq 48$; their sum at $(32, 48)$ is $2^{-117} + 2^{-25} < 1$. Hence the whole divisible product is zero, and its other factors are positive. Thus $X = 2^R$; the same universal projection then gives integral positive $\kappa = K/H$.

Now $q \mid X$ implies $q = 2^a$, $a \geq 5$. Equation (7), with $d = 0$, finally gives $q^6 \mid e_*$, because three is a unit modulo q. Thus ρ is integral. The carrier says $\sigma - WB/h$ is integral. Multiplying by q^2h and reducing modulo q^2 gives $q^2 \mid s$. Hence $q^3 \mid Y$, σ is integral, and $h \mid WB$. Since $W \equiv 1 \pmod h$, $h \mid B$, so t is a nonnegative integer.

The fixed-minus parity, exact half-binomial rounding, and fresh original native reconstruction now have their full retained premises. That source uses base $Y(X + 1) + 2$, not the auxiliary base A. The unchanged whole-tuple calibration produces a finite ZFC refutation.

1.5 Constructive positive completeness

A finite refutation supplies the retained canonical outer data $q = 2^a$, $X = 2^R = qw$, $R > 3q$, $R \equiv 3 \pmod 4$, $q^3 \mid Y$, $C > 0$, and $t = B/h \geq 0$ integral. Set $s = Y/q > 0$, and form

$$c = \psi_A(R), \quad k = 2\psi_{1+2XY^2}((R + 1)/2), \quad \tau = \chi_{1+2XY^2}((R + 1)/2), \quad \eta = c - kY, \quad \zeta = k - \eta.$$

The exact ratio theorem gives $0 < \eta < k$, hence both last inputs are positive. Equation (7) and dyadic q give integral ρ. Projection gives positive integral κ, and $\sigma = Y/q^3$ is integral.

A positive period of the Pell matrix modulo $c^2\tau$ gives a seed (f_0, I_0) with $c^2\tau \mid I_0$. Set $i = I_0/(c^2\tau) > 0$. Then $a = \Delta I_0^2 = f_0^2 - 1$, $F_0 = f_0^2$, and $G = (\tau f_0)^2$. The doubled pair $f = 2f_0^2 - 1$, $I = 2f_0I_0$ has exactly the required definitions. The retained auxiliary-completeness theorem gives positive V, y and the positive integer $T = (V + c + Rf^2)/(cf^2)$, the actual unshifted input.

The new ρ may be negative; the old nonnegative-quotient argument is not reused. Since $k \geq R + 1$ and $vk^2 > 6$,

$$|\rho| \leq \frac{k - R - 1 + vk(k^2 - 4)/6}{q^6} < \frac{k + vk^3/6}{q^6} < vk^3/3.$$

For every permitted $i \geq 1$,

$$g > \sqrt{\Delta}\, c^2E > c^2v^2k^2 > vk^3,$$

so $|\rho| < g/3$. Also $\sqrt{\Delta} > A - 1 = Y(X + 1) > X$ and $cE > 4$, hence $g > 4Xc$. As $\kappa < Xc$ and $\sigma \leq Y < Xc$, we have $\kappa + \sigma < g/2$. For either positive weight put

$$n = gy + (1 + hL)t + \rho - \kappa - \sigma.$$

This is integral, and $y \geq 1$, $t \geq 0$ imply $n > g - g/3 - g/2 > 0$. Its carrier is $z = gy$, so $P = 0$. All ten independent coordinates have been constructed strictly positively; no inverse positive shear or unproved sign is used. Every permitted period seed works, without additional cofinal enlargement.

1.6 Exact costs and verification

Deleting the incoming gate $v = (XY)Y$, restoring the sharing $\ell = (XY)(kY)$, and replacing its four-gate rank cubic by k^2, $k^2 - 4$, $\ell(k^2 - 4)$ saves two multiplications. The principal has 51 multiplications, 18 additions and 16 subtractions. All inputs, ports and rows are live.

Put $q_0 = \beta J$, $k_0 = \eta + \zeta$, $C_0 = q_0 - F - \alpha$, and $B_0 = k_0 sT - q_0^2(q_0 - F)$. The *complete* highest homogeneous forms are

$$[P_{bV}]_{304} = -9 \cdot 2^{12} q_0^{128} w^{34} s^{78} i^{14} k_0^{44} B_0^2,$$
$$[P_{V^2}]_{410} = 9 \cdot 2^{16} q_0^{166} w^{46} s^{104} i^{20} k_0^{60} B_0^4 C_0^2.$$

Their full expansions have respectively 274 and 3750 terms. Nonzero coefficients at maximal T-power establish the exact degrees for the actual $\beta > 0$, not merely a formal upper bound or a ray sample.

The producer-independent checker covers all actual rows with 37 exact free-polynomial cuts per graph, verifies full leading forms and every coefficient byte, reconstructs the four fixed numerals, and rejects both alternative operations at every gate plus a missing-correction mutation: 171 rejections for the principal and 169 for the control. Separate finite recurrence controls include the obstruction at three; they do not replace the unbounded arithmetic proof. Normal and optimized Python receipts agree byte for byte.

```
python3 -B positive/build.py --check
python3 -B positive/verify.py
python3 -O -B positive/verify.py
```

1.7 Whole complemented-label radix47 sidecar

The same two free-port row lists are separately specialized to the *entire* frozen complemented-label radix47 tuple. The principal remains 85/304, but its largest fixed port is 415,904 bits and the four raw numerals occupy 207,214 bytes rather than 295,388. This reuses the incoming encoding and controller; it is not a new radix theorem and is not the repository's conditional three-anchor radix47 layout. The canonical radix67 equations remain separate.

The new arithmetic is uniform in the same scalar, source-shift mask and population interface already checked for the whole47 tuple. In soundness it recovers $C > 0$, $X = qw = 2^R$, $q^3 \mid Y$, the native half-binomial, and $(K_0 + w)C - F = t(q - 1)$, $t \geq 0$. These are exactly the retained whole47 bridge's premises, with $W_{\text{native}} = w$ and source transport variable $t + 1$. Its fresh Pell witnesses are constructed at $Y(X + 1) + 2$, before the actual complemented-label decoder is applied. Conversely, the actual whole47 controller supplies dyadic q and the same outer interface; choosing $w = W_{\text{native}}$, $s = Y/q$, and applying the explicit positive construction above gives a new zero. No old67 decoder or controller is silently substituted.

The complete ordinary native bridge, incoming P66 equation, all four payloads, 3297 coefficients, 51 native target identities, and its 37 pinned dependencies are retained under `positive/radix47/inherited/`. The sidecar checker requires identical free-port rows and an exact whole-tuple binding, reconstructs every numeral, and repeats the cuts, leading-form and mutation checks. Its normal and optimized receipts agree. The default main checker includes this sidecar; its literal binding can also be reproduced independently:

```
python3 -B positive/radix47/build.py --check
python3 -B positive/radix47/verify.py
```

2 Complete fixed-coefficient recipe

The following finite recipe is retained from the inspected ProveIt indexed constant specification, whose exact source and SHA-256 are recorded in `inherited_lucas/parent/proof_sources/indexed_constants.tex` and `SOURCE_PINS.json`. It defines the entire unchanged BFS943 tuple, not a new specialization. The symbol K in the inherited recipe is renamed K_0 to agree with Section 1; symbols in this section are local fixed data, not additional unknowns. The supplied standard-library reconstruction checks every resulting byte against both literal circuits. The source shift is already included in M_F. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\begin{aligned}\Phi(x) &= (1 + 2^{51}) G_{943}(x) - 2^{10} \sum_{i=0}^{942} v(t_i) x^i \\ &\quad + \sum_{(b,e,m) \in \mathcal{C}} 2^b x^e G_m(2x), \\ \mathcal{C} &= \{(10, 1023, 8), (18, 1013, 9), (27, 1003, 8), \\ &\quad (35, 983, 9), (44, 1031, 1), (45, 1022, 1), (46, 1011, 1), \\ &\quad (47, 992, 1), (48, 1095, 1), (49, 2195, 1), (50, 3296, 1)\}.\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$\begin{gathered}h(q, b) = 2(q + 1) + b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\}, \\ \mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\}, \\ r_q = 6 + |\mathcal{R} \cap \{0, \dots, q - 1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0, \dots, q - 1\}|.\end{gathered}$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are

fixed:

index i	tile $t_i = (w, n, e, s)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j + 2\ (j = 2q + b < 430)$	$((1 - d_j) l_{u_j}, h(q, b), d_j r_{u_j}, v_j)$
$432 + b\ (b = 0, 1)$	$(0, h(215, b), 0, h(215, b))$
$434 + b\ (b = 0, 1)$	$(0, b, 0, b)$
$424 + 2r_q + b\ (q \in \mathcal{R}, b = 0, 1)$	$(r_q, b, 0, h(q, b))$
$424 + 2l_q + b\ (q \in \mathcal{L}, b = 0, 1)$	$(0, b, l_q, h(q, b))$
$924 + j\ (0 \le j < 4)$	$(0, j + 430\lfloor j/2 \rfloor, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934+3a+b-6\lfloor a/4 \rfloor$ $a \in \{0,1,4\},\ b=0,1,2$	$(a, 437, b, 437)$

For $t = (w, n, e, s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\, 2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A, k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430 + q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1 - m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1 + \sigma_j) + (1 - \nu_j) D_{216}(T_0, j - \sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b + 1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

3 Complete complemented-label radix47 constants

This is the entire incoming complemented-label tuple, not the repository's conditional three-anchor radix47 layout. All indices, labels and sums below are fixed data. The finite recipe and 436-character seed are reproduced from `radix47/inherited/provenance/radix47/radix47/equation.tex`; its K is renamed K_0. The independent reconstruction checks all 943 tiles, all 3297 coefficients, and every byte of all four actual fixed integers. No new encoding is claimed.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{8807}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{4393} + \rho^{5495}.$$

$$\begin{aligned}
M_C &= \beta + \rho - (\rho+3)\rho^{1063} \\
&\quad - \sum_{e=0}^{3296} \lfloor (D_\rho(\Phi(\rho), e) + \rho - 1)/\rho \rfloor \, \rho^e, \\
M_F &= 2B + 3 + (2^{47} - 2)\Lambda \\
&\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\
&\quad - \rho^{8792} + (2^{28} - 3)\rho^{7694} - \rho^{7691}, \\
K_0 &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\
&\quad + \Lambda\Phi(1/\rho),
\end{aligned}$$

where

$$\begin{aligned}
\Phi(x) = \sum_{i=0}^{942} &(1 + 2^{10} + 2^{10} w_i + 2^{18} n_i \\
&+ 2^{27} p_i + 2^{35} s_i) x^i \\
&+ 2^{10} x^{1003} (x^{20} + 2^{17}) G_8(2x) \\
&+ 2^{18} x^{983} (x^{30} + 2^{17}) G_9(2x) \\
&+ 2^{44} x^{992} (1 + x^{19} + x^{30} + x^{39}) \\
&+ 2^{44} x^{1095} (1 + x^{1100} + x^{2201}).
\end{aligned}$$

The rational intermediate in K_0 cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \ldots, q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are

fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
$0, 1$	$(2, 435, 3, 1),\ (5, 0, 0, 0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$434 - 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 432a+b)$
$\substack{424+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$924 + 2a + b\ (a, b = 0, 1)$	$(0, 432a+b, 0, 438)$
928	$(2, 434, 2, 0)$
929	$(3, 436, 4, 4)$
930	$(4, 436, 4, 0)$
$497 + t\ (t = 434, 435, 436)$	$(1, 438, 1, t)$
$934 + 3a + b\ (a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
OOe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```